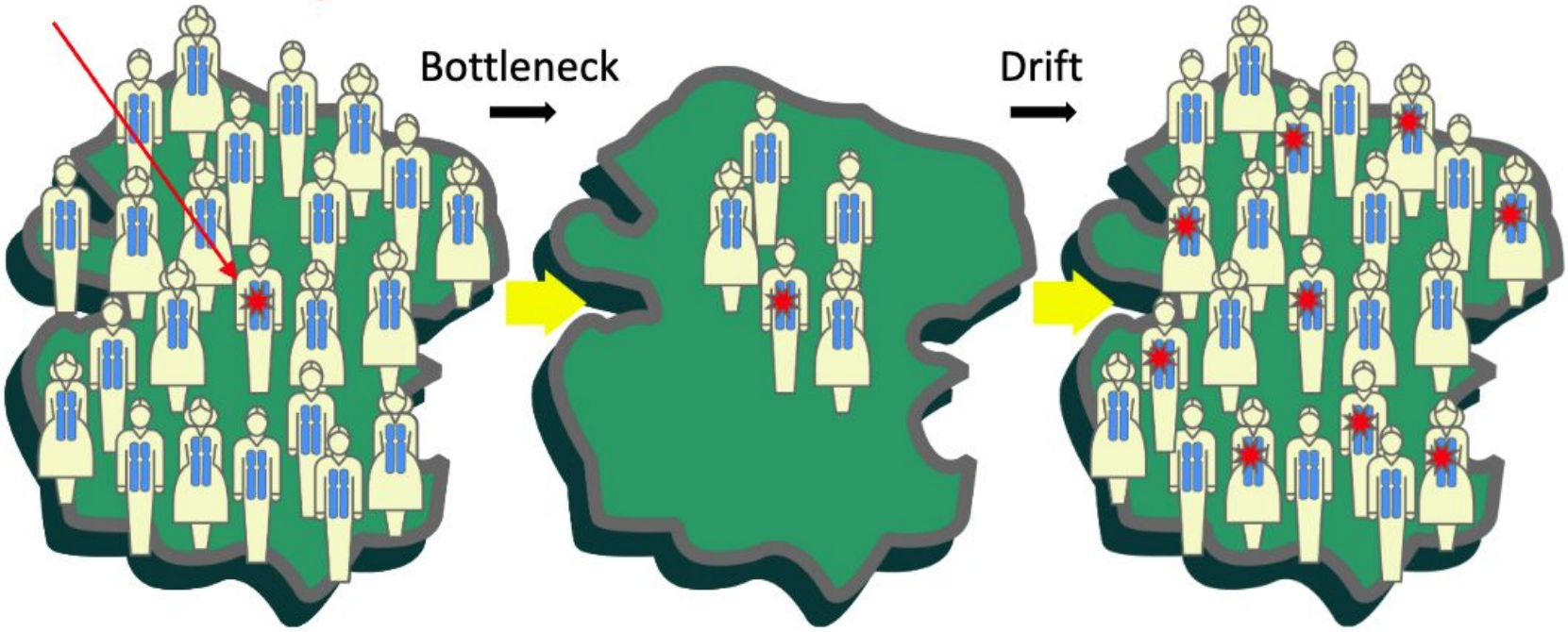
Dog Genetics

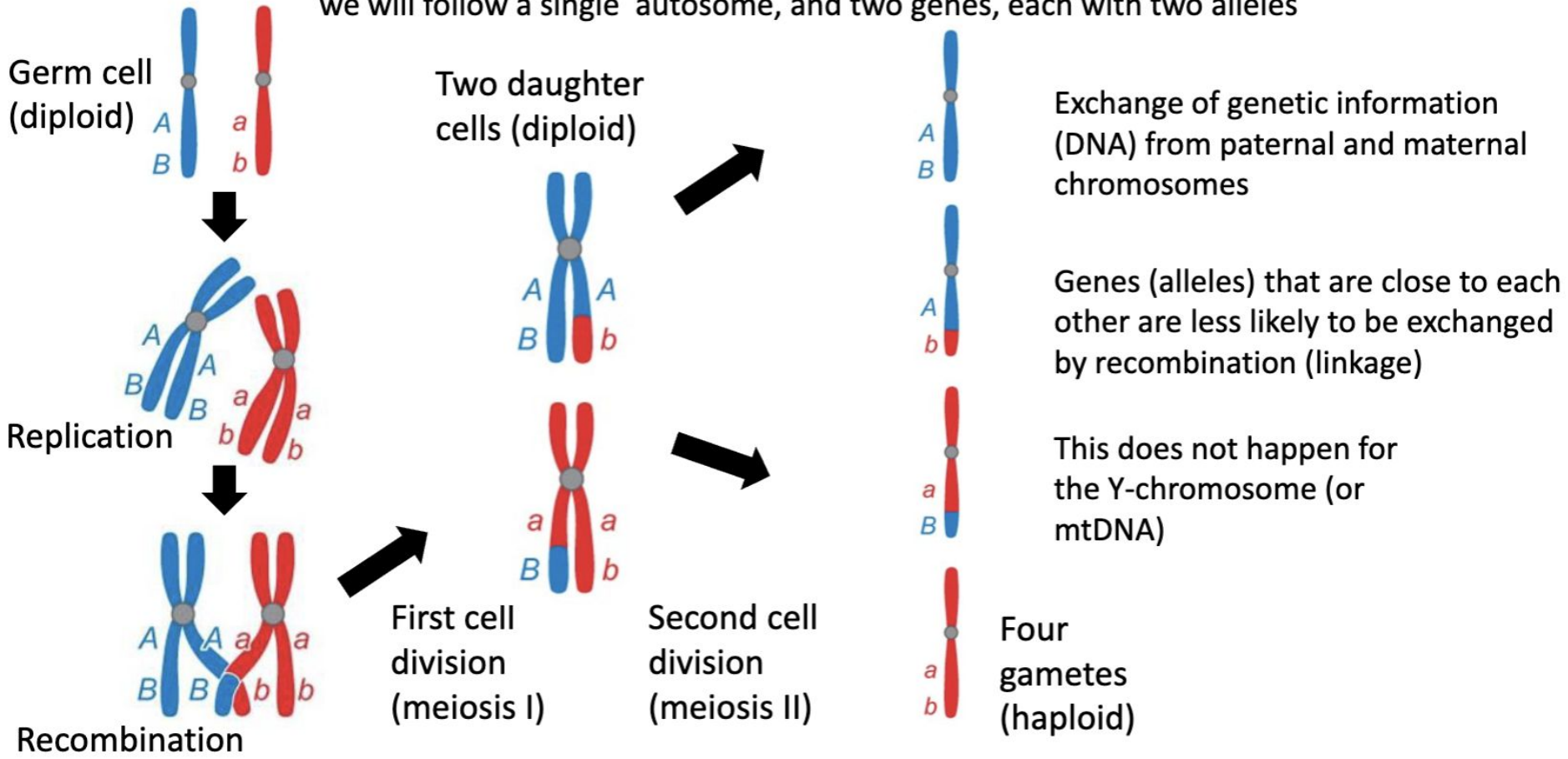
Erin Wissink, PhD
April 6, 2026

Population bottlenecks



DNA Recombination

Meiosis – specialized cell division of germ cells, which produces gametes (egg, sperm)
we will follow a single autosome, and two genes, each with two alleles



One species with extraordinary differences in age, size, and shape



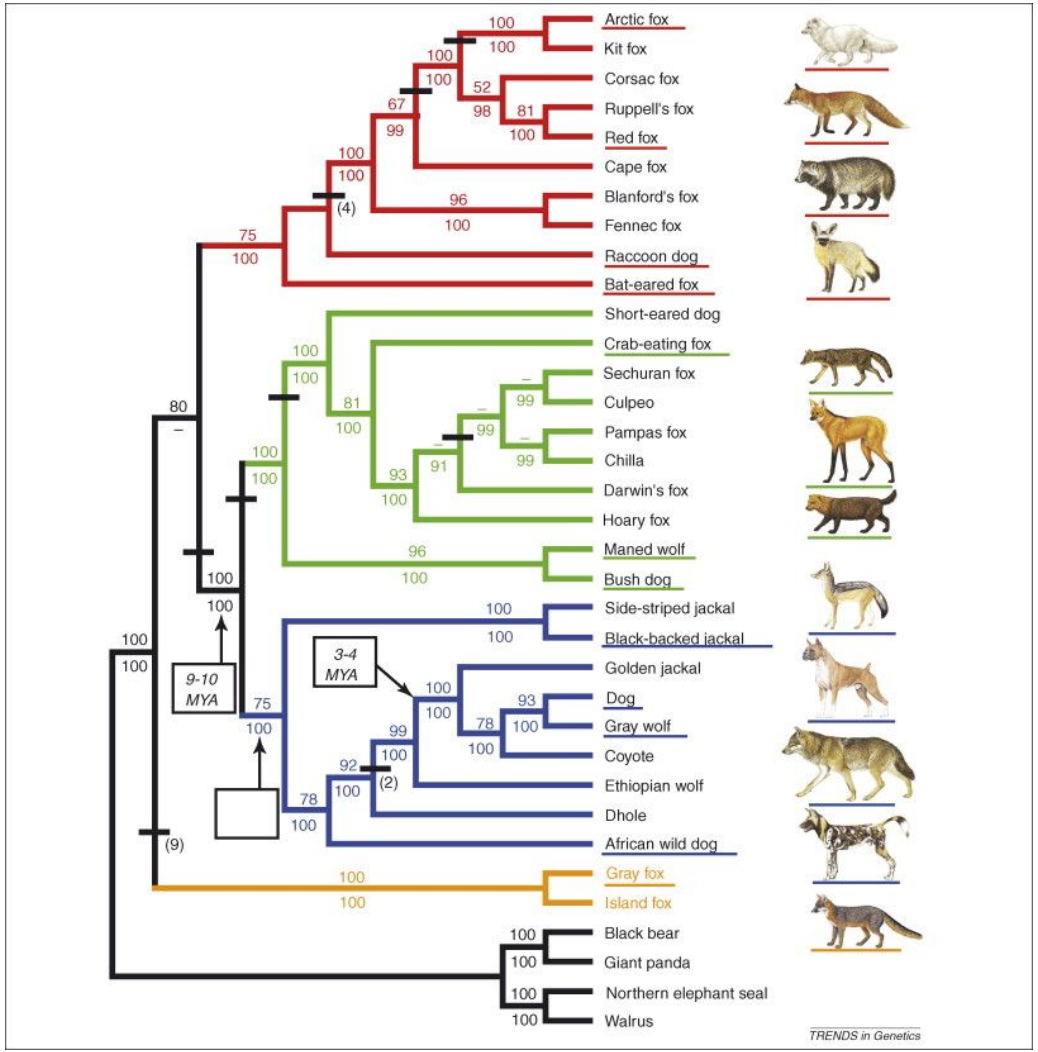
Dogs range from 5 to 200 pounds at full growth, with shoulder height from 5 to 36 inches

Small dogs can live 20 years, whereas largest dogs live ~8 years

What has caused this amount of diversification within one species?



https://ssec.si.edu/stemvisions-blog/your-best-friends-genes



Modern canids originated ~ 50 million years ago

See wolf-like canids 6 million years ago

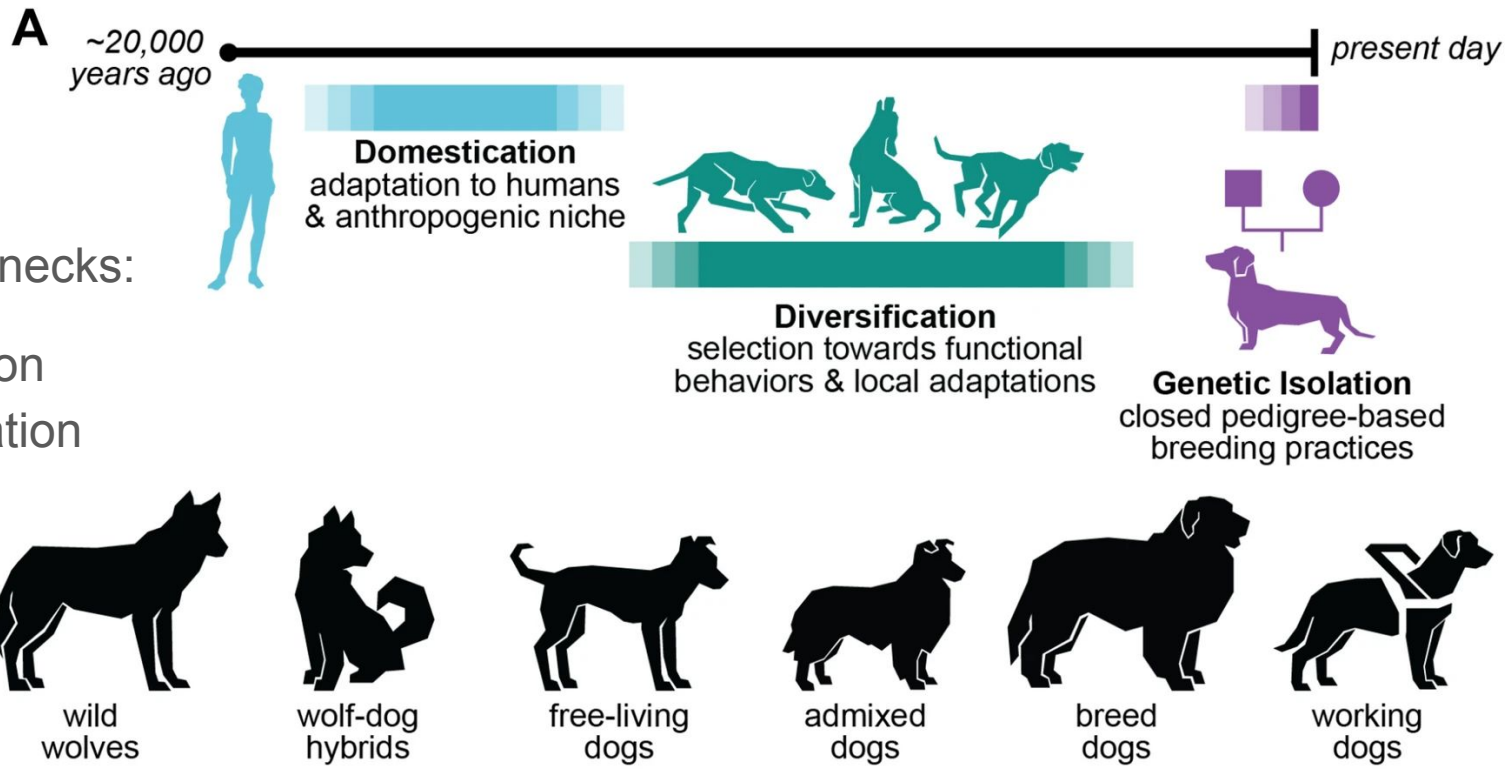
Dogs and wolves are still able to interbreed and produce viable, fertile offspring

Wayne & Ostrander, Trends in Genet (2007)

Wolf to dog evolution and bottlenecks

Two major bottlenecks:

1. Domestication
2. Breed formation



Morrill et al, *Human Genet.* (2023)

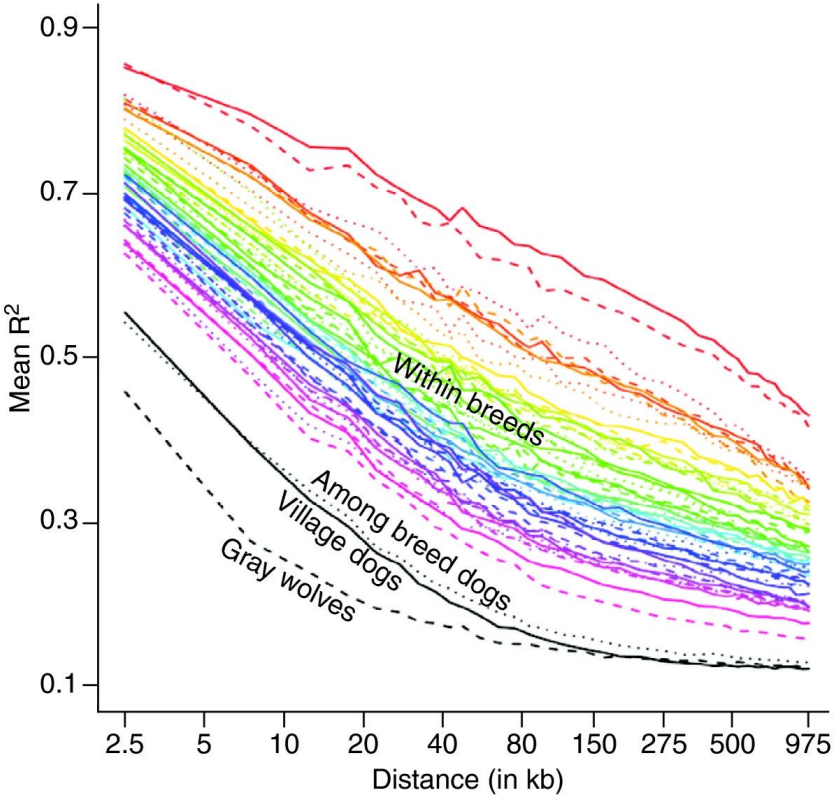
Linkage disequilibrium in dogs

What can linkage disequilibrium tell us about evolution?

In humans, LD extends for kilobases (thousands of bases)

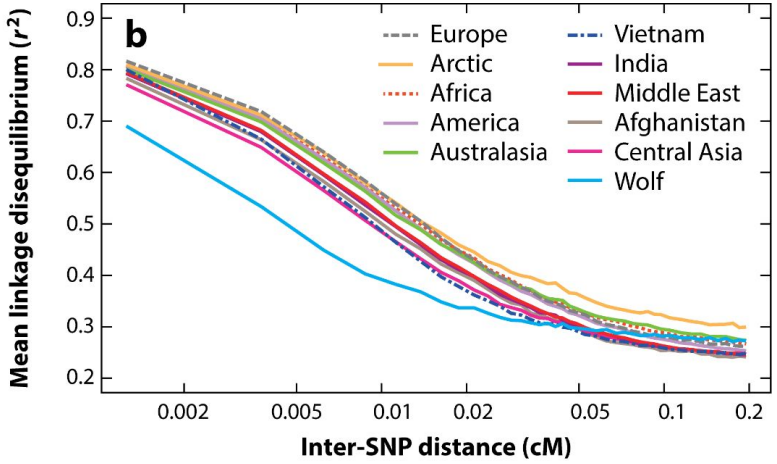
Gray wolves are similar

In purebred dogs, stretches are much longer -- sign of inbreeding!



Boyko, *Genome Biol* (2011)

Village dogs genetic analyses suggest dogs evolved in central Asia





Can use linkage disequilibrium to measure genetic diversity

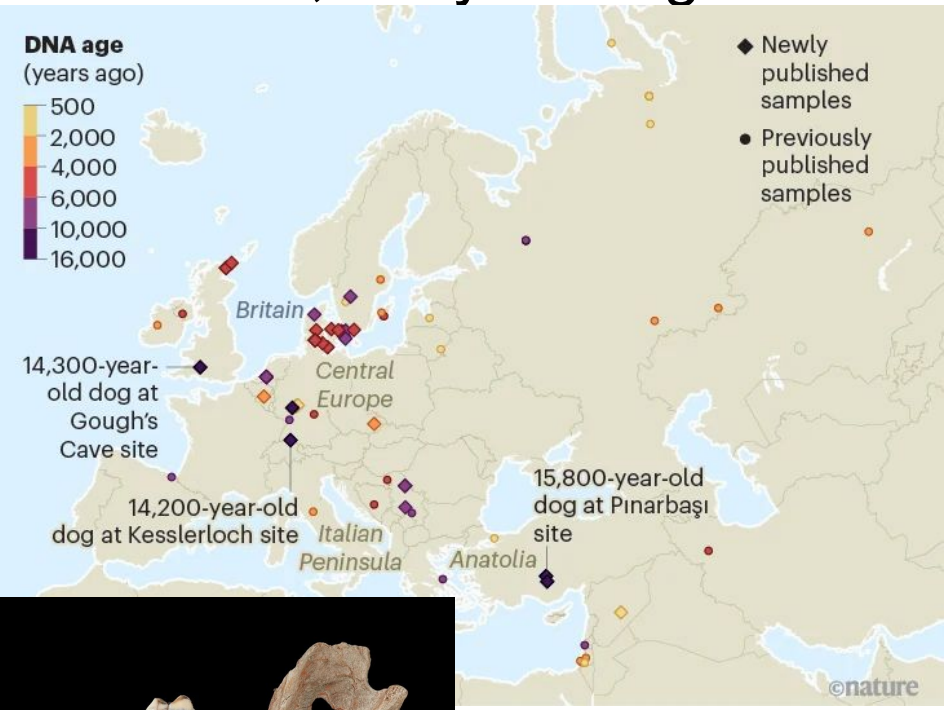
Village dogs are free-living populations

Village dogs have longer LD than dogs because of the domestication bottleneck but shorter than breed dogs

LD from dogs in central Asia (Nepal and Tibet) is shorter than from other village dogs

https://embarkvet.com/resources/it-takes-a-village-dog/; Freedman & Wayne, *Ann Rev Animal Biosci* (2017)

Ancient DNA analyses show dog domestication occurred at least 14,000 years ago





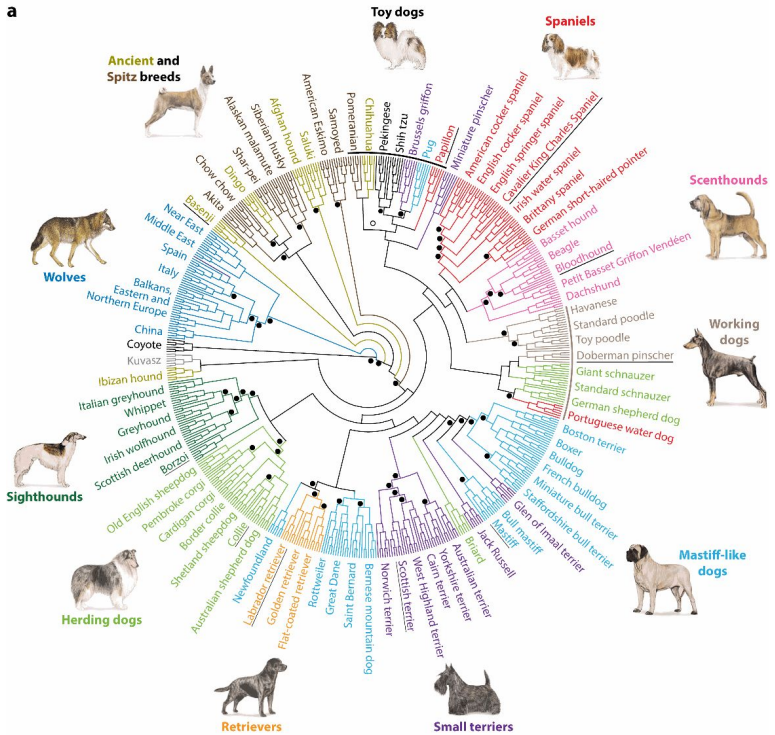
Dogs in Anatolia (Turkey) 16K years ago were highly related to dogs in Great Britain (14K years ago)

Human populations at that time were more diverse -- did human cultures pass on dogs to each other?

The ability to extract DNA from ancient remains gives hints into evolution, human and dog

https://www.nature.com/articles/d41586-026-00378-2

Second population bottleneck occurred when breeds were created



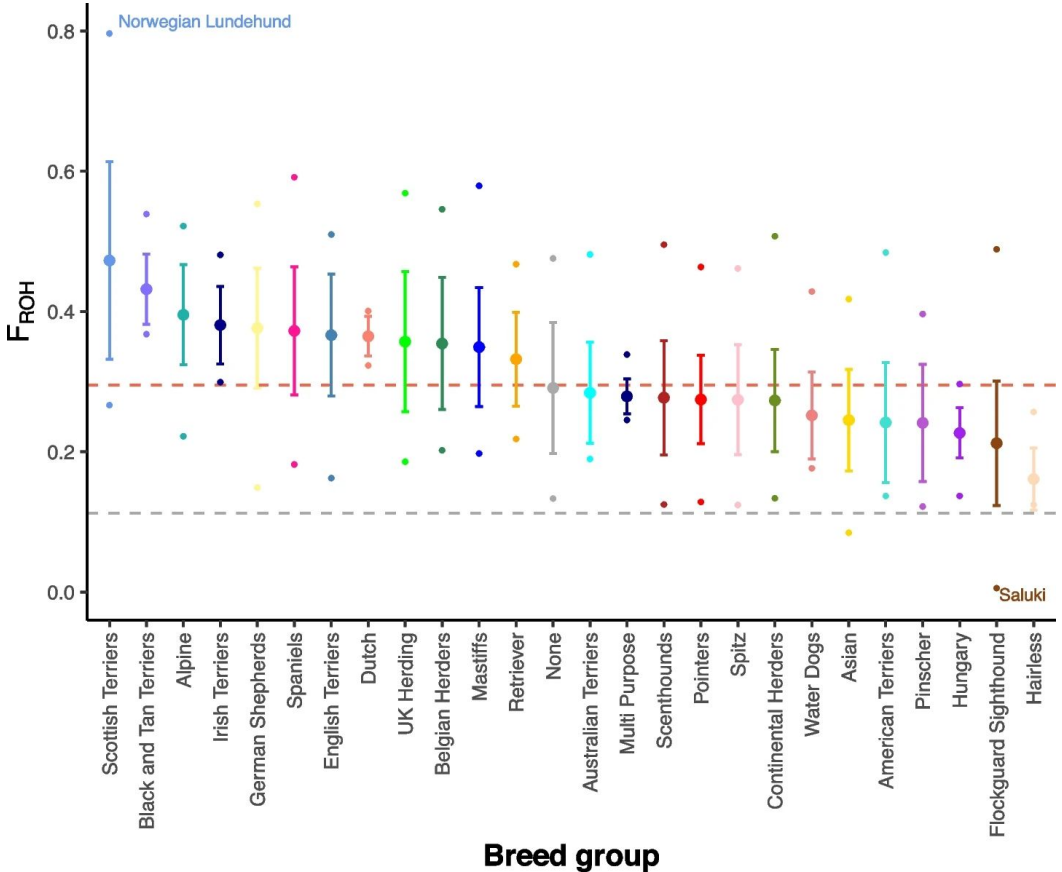
Dogs had phenotypic differences due to geography and being bred for different functions (herding, hunting, etc)

The Kennel Club was the first breed club, starting in 1873. They set the first breed standards and closed stud books

Inbreeding levels increased, bringing along health problems

Freedman & Wayne, *Ann Rev Animal Biosci* (2017)

Inbreeding levels in different dog breeds



Homozygote: both alleles have identical sequence

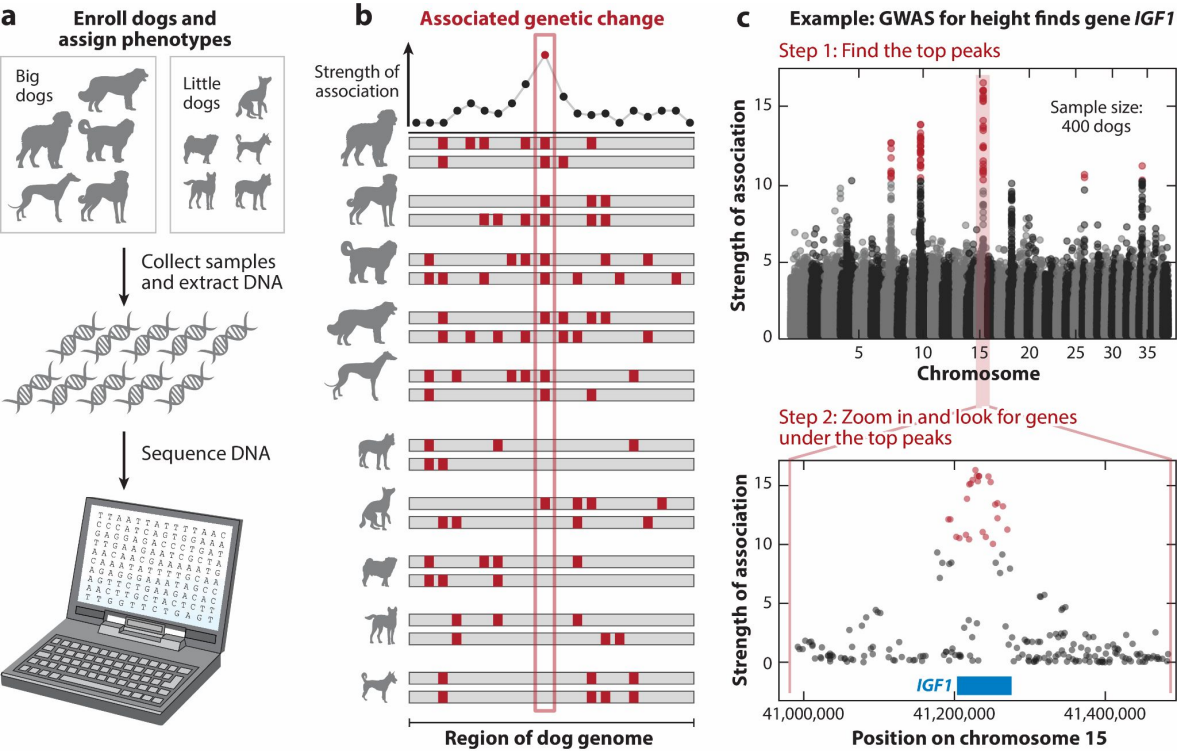
Run of homozygosity: Long stretches of identical sequence (haplotypes) indicates recent inbreeding

F_{ROH}: Fraction of genome with runs of homozygosity

Gray line shows mean F_{ROH} for wolves

Meadows et al, *Genome Biol* (2023)

Finding variants that explain trait diversity in dogs



In humans, some traits are Mendelian, but these are usually rare (sickle cell disease, Tay Sachs disease)

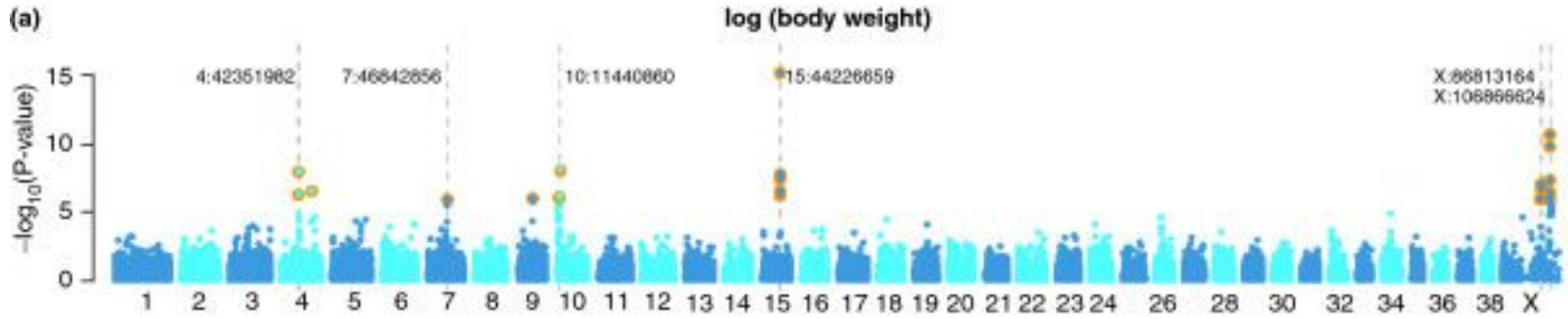
Most traits like height are associated with 1000s of variants in the genome

What about dogs?

Lord et al, *Annu Rev Anim Biosci.* (2025)

Lord KA, et al. 2025
Annu. Rev. Anim. Biosci. 13:167–88

Size from handful of loci vs 1000s for humans

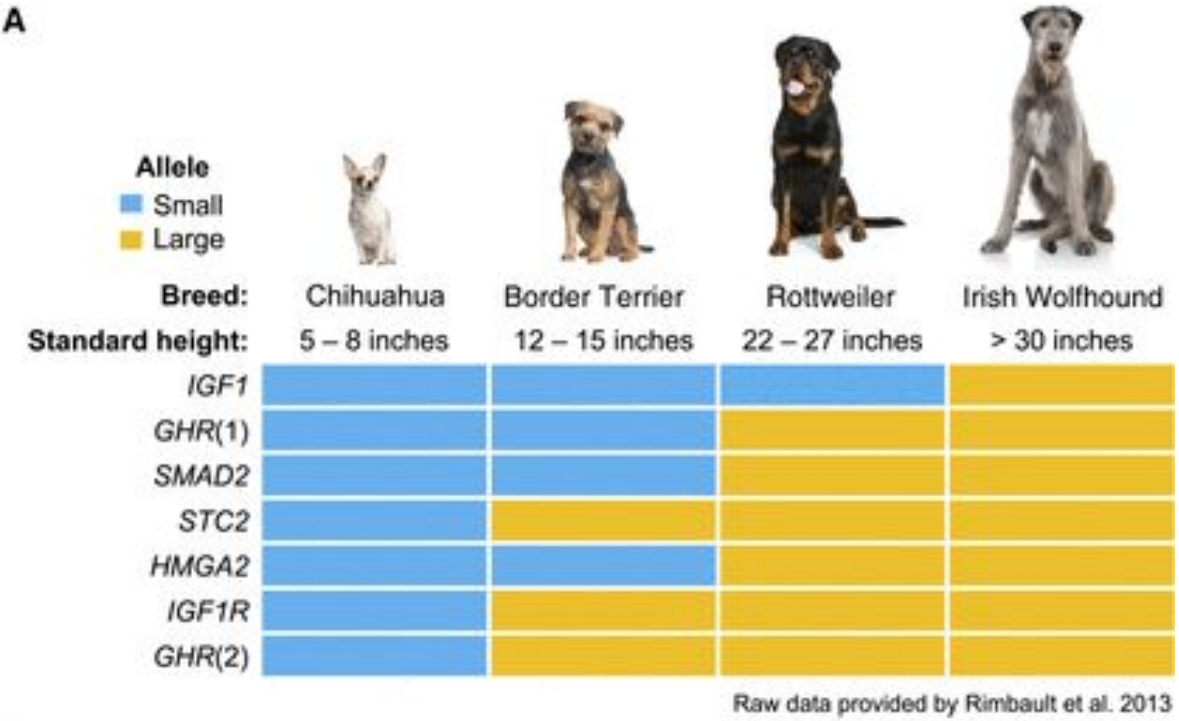


Manhattan Plot from GWAS on dog body weight shows 6 significant associated sites

Boyko, *Genome Biol* (2011)

Sites that explain dog size variability

A



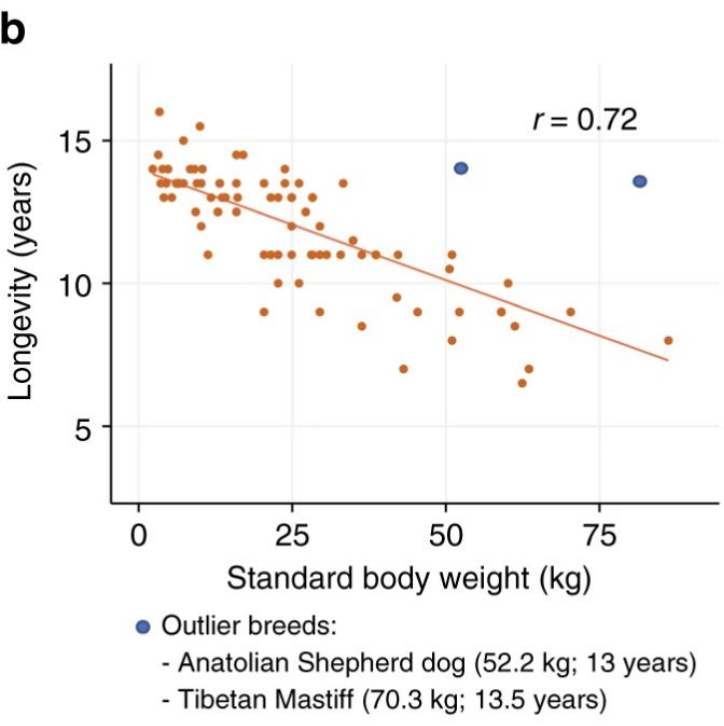
50% of variation in dog size is explained by only 7 sites in the genome

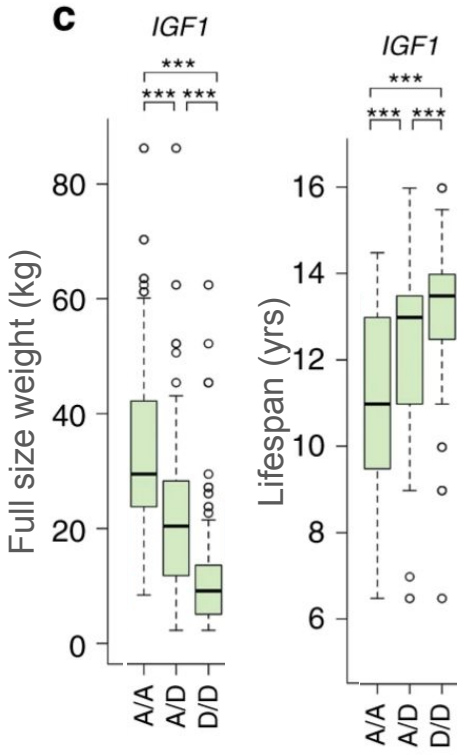
IGF1 and IGF1R are in the insulin signaling pathway

GHR is the growth hormone receptor

Buckley & Ostrander, *Genome Res* (2024)

Size and lifespan are inversely correlated in dogs





SNP in IGF1:

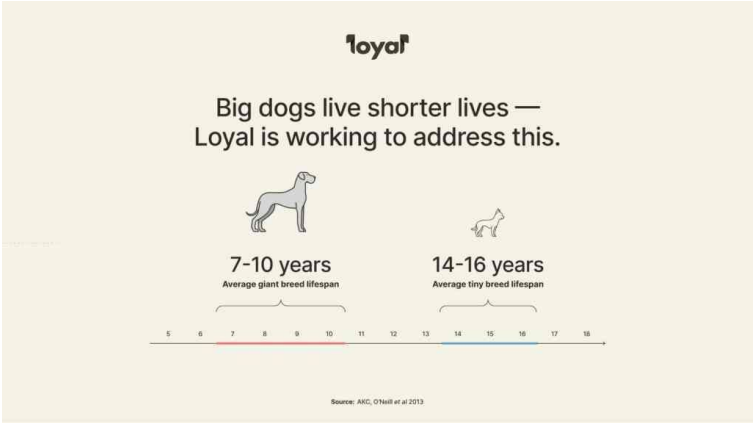
A/A is found in large dogs with short lifespans

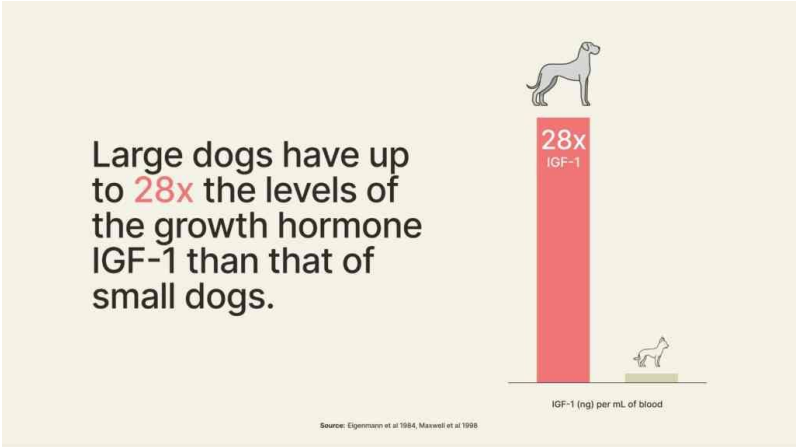
A/D is found in medium dogs with medium lifespans

D/D is found in small dogs with long lifespans

Plassais et al, *Nat Comm* (2019)

Targeting insulin signaling to extend lifespan in big dogs

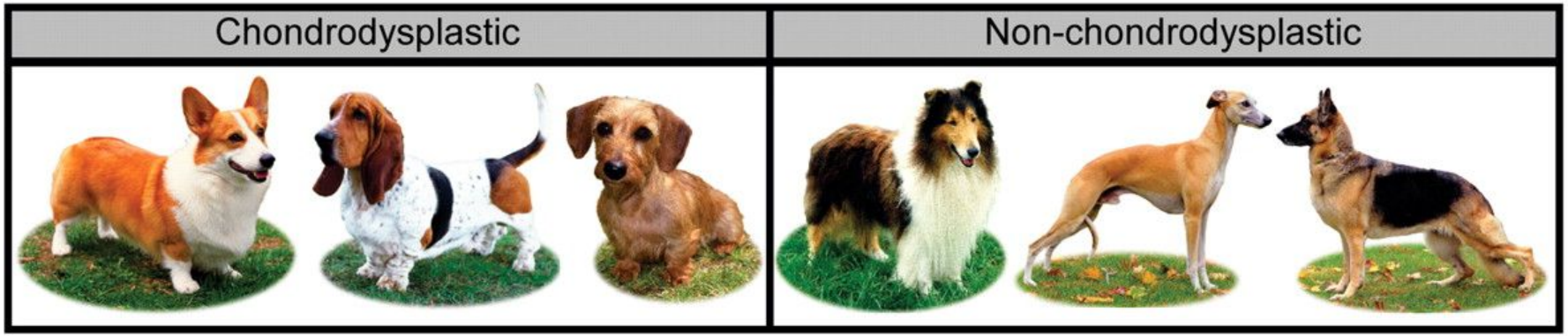




LOY-001 extends lifespan in part by reducing IGF-1 to levels seen in smaller-breed dogs. The IGF-1 axis is one of the most well-studied longevity pathways. In model organisms from *C. elegans* to mice, reducing IGF-1 extends healthy lifespan, and increasing IGF-1 shortens healthy lifespan. In humans, certain centenarians have been shown to have genetically lower levels of IGF-1.

https://loyal.com/posts/loyal-announces-historic-fda-milestone-for-large-dog-lifespan-extension-drug

Mutation for short dogs



Chondrodysplasia: disproportionately short legs with a long body

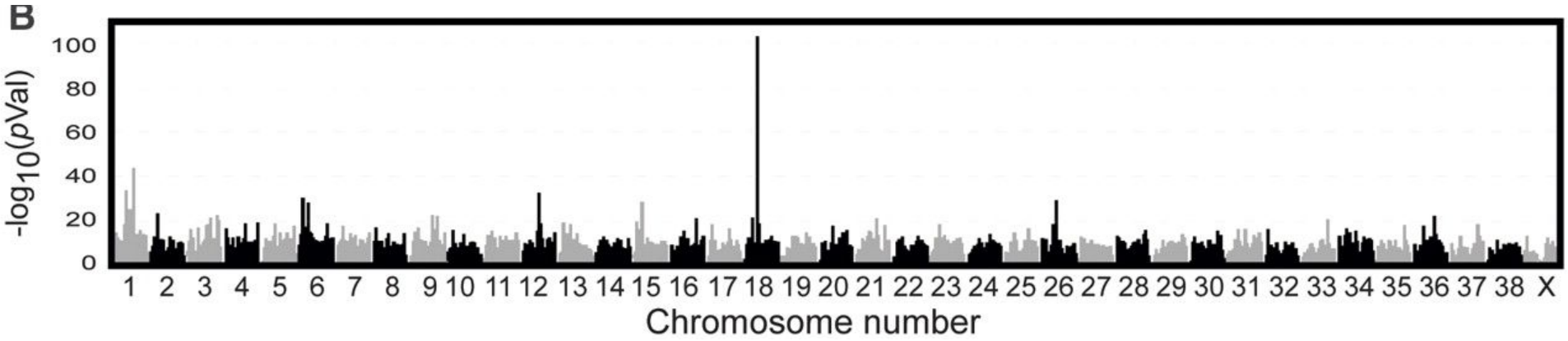
Found in corgis, basset hounds, Pekingese, Dachshunds

Shortness has incomplete dominance -- one copy of the variant allele has an intermediate phenotype

What makes their legs short?

Parker et al, *Science* (2009)

Short dog mutation is caused by a retrogene



GWAS showed a significant hit on chromosome 18

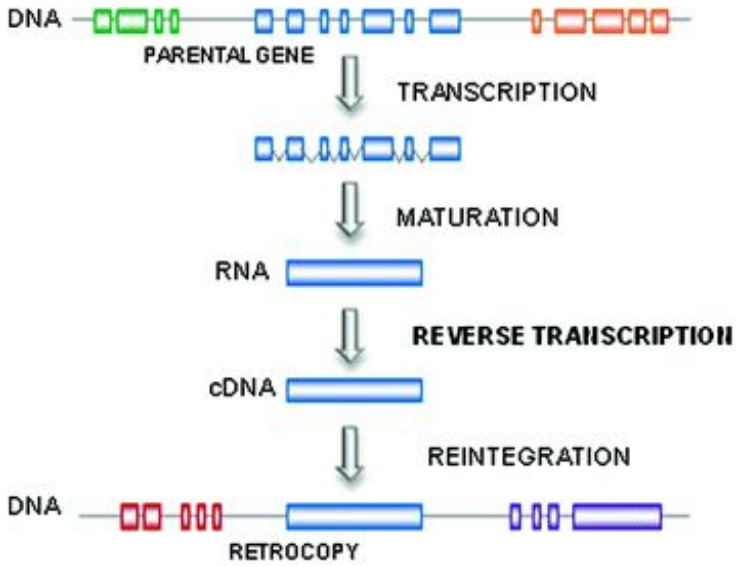
In short dogs, found a 5kb insertion -- it was an extra copy of the FGF4 gene from a retrogene!

FGFR3 is found in human dwarfism. The extra FGF4 excessively activates FGFR3 in dogs, which calcifies the growth plate early in development

Parker et al, *Science* (2009)

Retrogenes





Barbara McClintock (Cornell 1923, 25, 27) discovered “jumping genes” that move around the genome and are responsible for corn coloration

https://www.nobelprize.org/stories/women-who-changed-science/barbara-mcclintock/

Several dog species have an extra copy of FGF4 that “jumped” into chromosome 18 after being transcribed into RNA, than reverse transcribed back to DNA

Ciomborowska et al, Evo Biol: Mech & Trends (2012)

Coat genetics overview

	PHENOTYPE	*FGF5*	*RSPO2*	*KRT71*
A	Short	-	-	-
B	Wire	-	+	-
C	Wire and Curly	-	+	+
D	Long	+	-	-
E	Long with Furnishings	+	+	-
F	Curly	+	-	+
G	Curly with Furnishings	+	+	+



Many different coat textures -- fur length, curliness, wiriness, furnishings -- are due to combinations of changes at just 3 genes in the genome

Cadieu et al. Science (2009)

Coat genetics: fur length

Many different mutations in the FGF5 gene cause long fur, a recessive trait

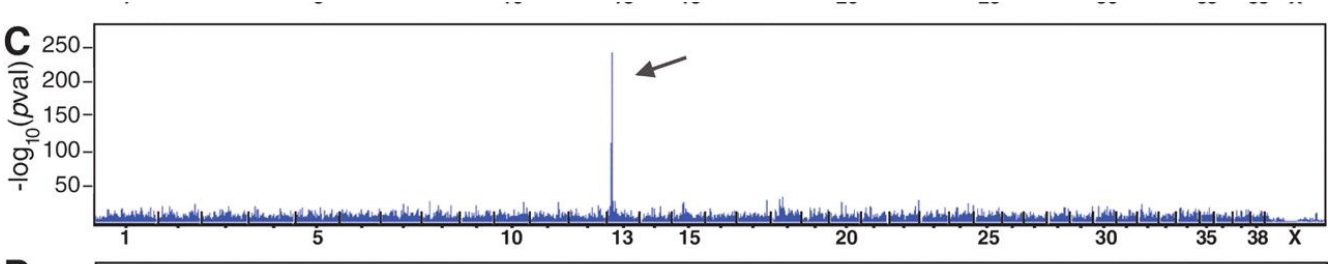
Many breeders have selected for long fur. The underlying variation is different but targets the same gene



Everts et al, *Animals* (2026)

Coat genetics: furnishings





Dogs can have “furnishings” aka eyebrows and mustaches and is the breed standard for Labradoodles, Goldendoodles, and Portuguese Water Dogs

Dominant trait based on a 167 bp insertion in the RSPO2 3′ UTR. This variant increases gene expression, which then creates longer hair follicles

Cadieu et al, *Science* (2009)
https://vgl.ucdavis.edu/test/furnishings-and-improper-coat

Coat genetics: curly fur

KRT71 is a keratin gene

Two different mutations in this gene cause curly fur in dogs: R151W (missense) and S422Rfs (frameshift)

Incomplete dominance: one mutation causes wavy fur, two mutations causes curly fur

Mutation in the same gene causes curly fur in cats, like Devon Rex







Cadieu et al, *Science* (2009)
Gandolfi et al, *Mammalian Genome* (2010)
https://vgl.ucdavis.edu/test/curl

What does breed tell us about behavior?

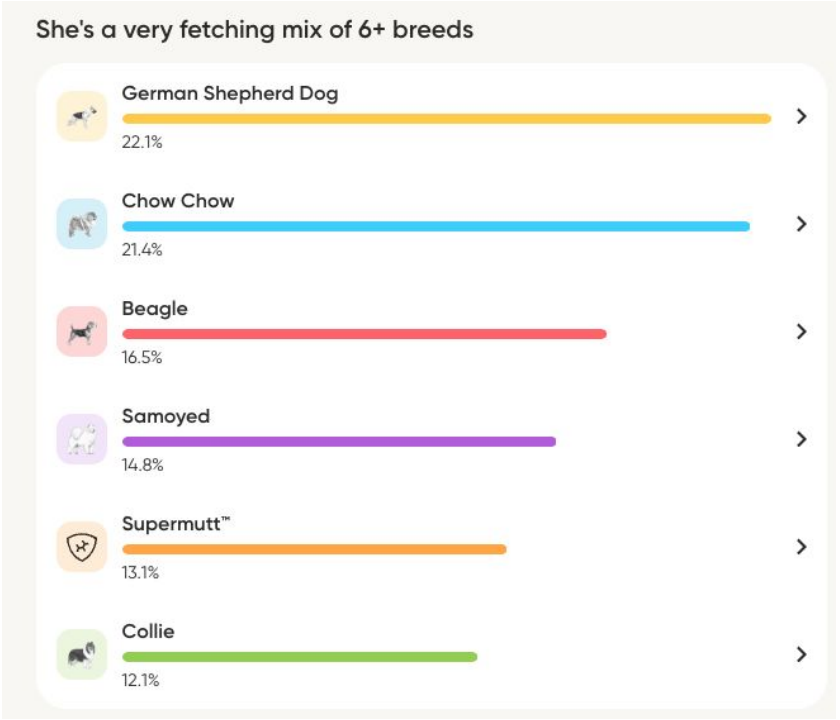


Some breeds have very “stereotypical” behaviors -- herding, pointing, soft mouths

These are traits that are ingrained that are honed with training

Meet Saki





We suddenly understood why our “Samoyed/Golden retriever” refused to play fetch!

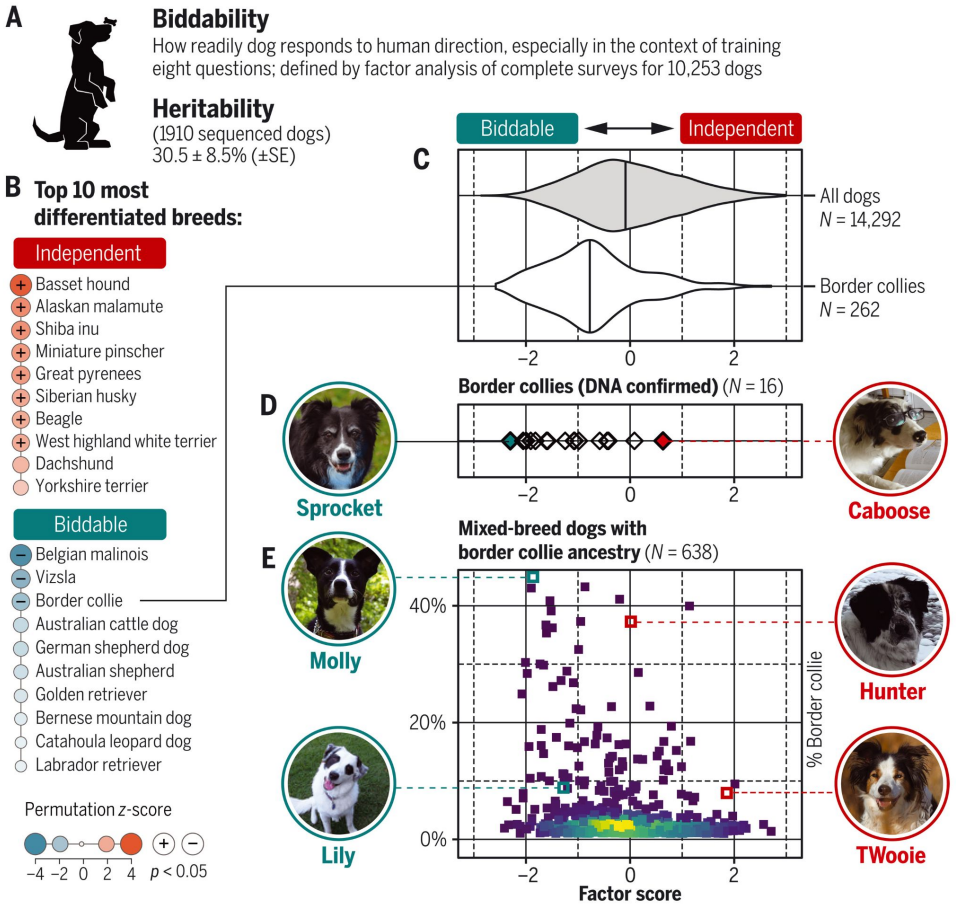
How much does breed tell us about personality?

Personality traits like biddability and human sociality are highly variable between and within breeds

~10% of a dog's personality can be explained by breed makeup

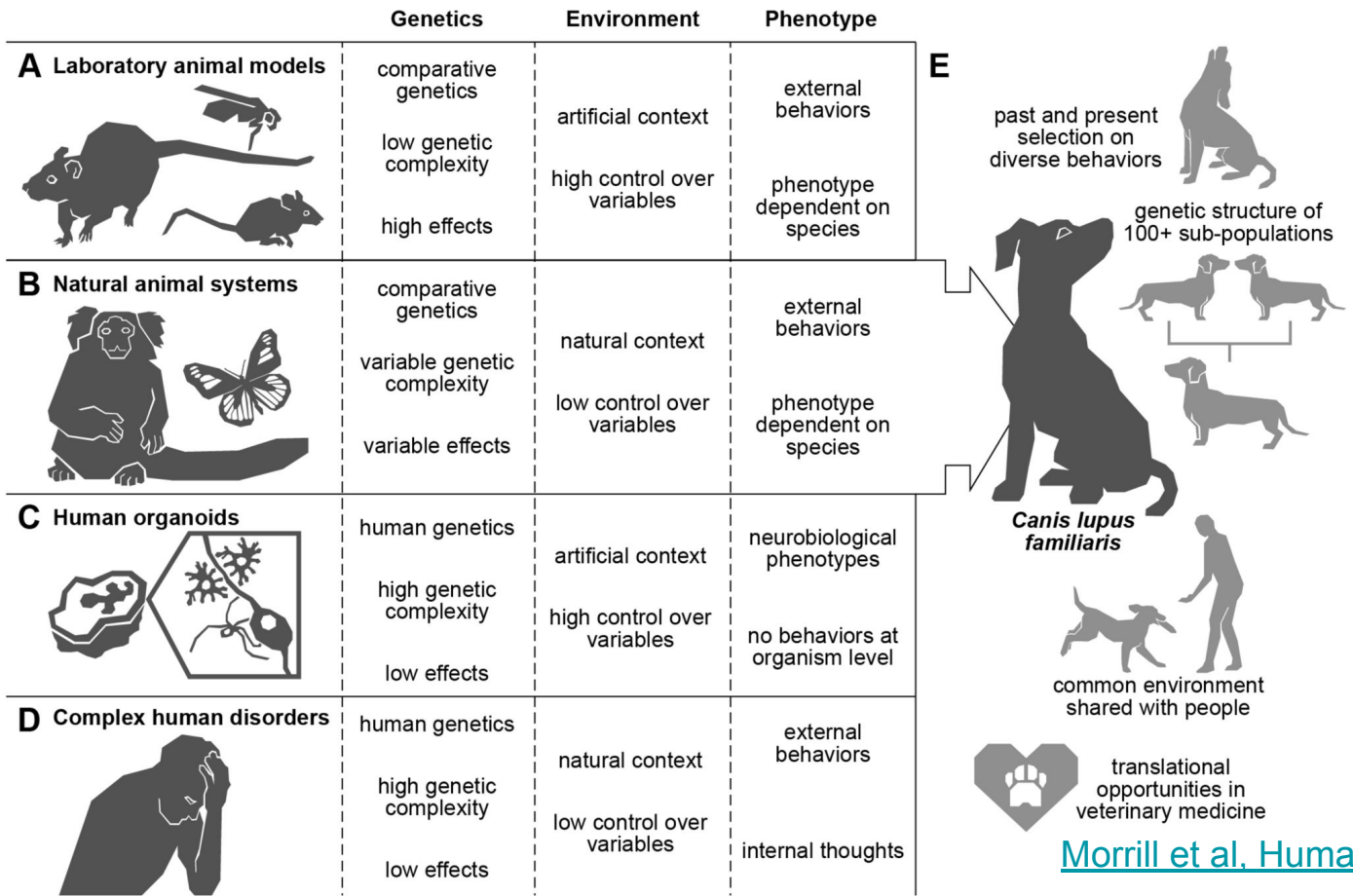
People with mutts will guess the breed makeup, in part based on personality

Environment usually has bigger impact than genetics



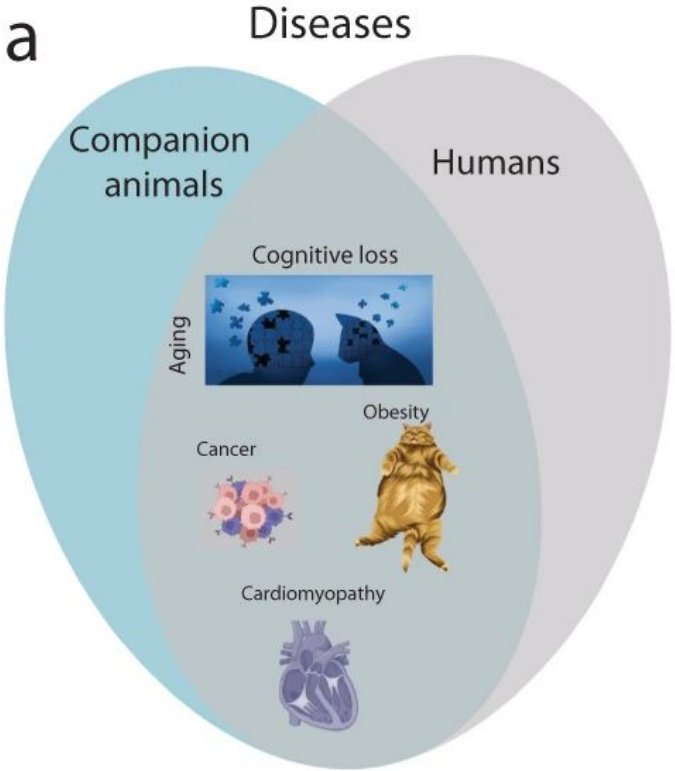
Morrill et al, Science (2022)

Dogs as a model for human disease



Morrill et al, Human Genetics (2023)

What can we learn from dogs?



Dogs provided info about narcolepsy and possible drug targets

Dog osteosarcoma informs human treatment

Cornell professor Dr. Jacquelyn Evans researches gastric cancer in dogs and how it can help human treatment

De Sousa et al, *Comm Biol* (2023).

Breeding issues: One mutation, two traits



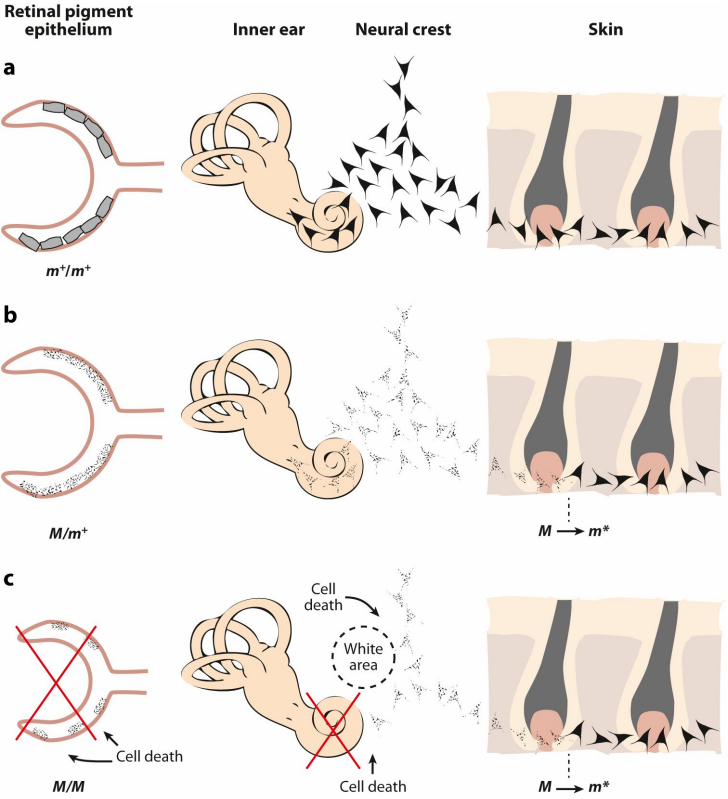
Merle coats are mottled and found in breeds including Australian shepherds and border collies

Trait is dominant and caused by a variant in the PMEL gene

Breeding two merle dogs is not recommended because double merles are often deaf

https://www.akc.org/expert-advice/dog-breeding/merle-in-dogs/

Why is deafness common in double merles?



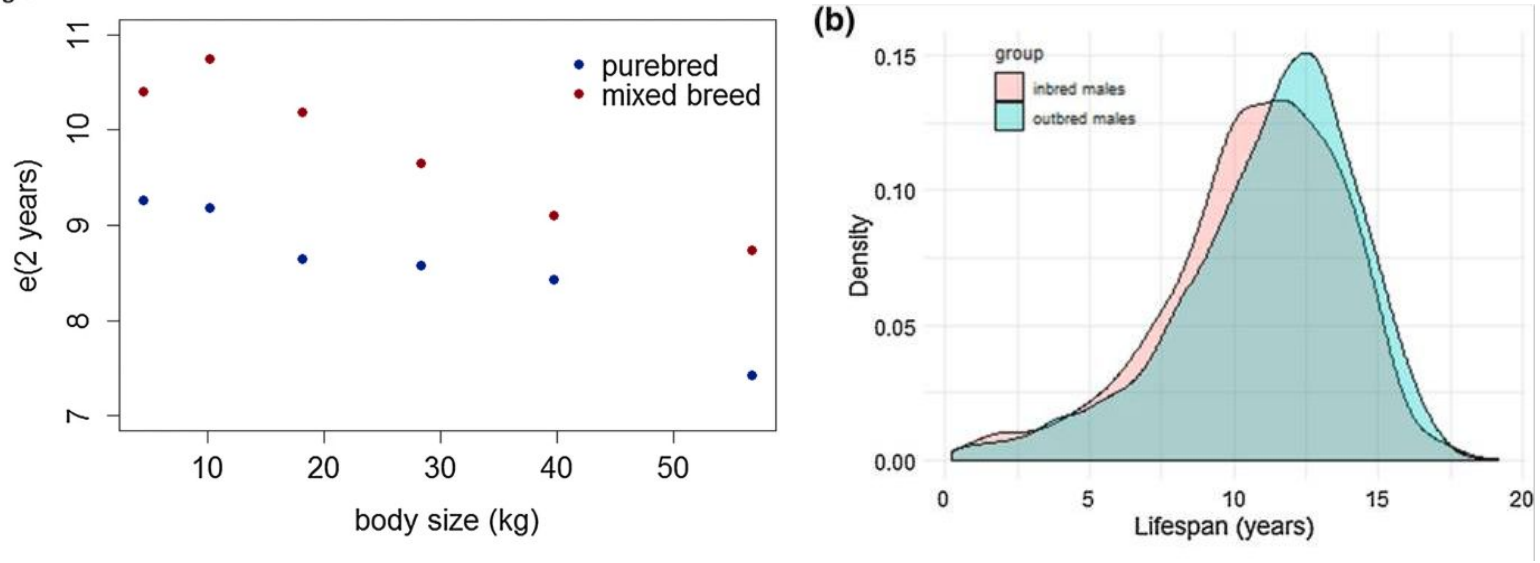
Kaelin CB and Barsh GS. 2013.

The merle mutation in the PMEL gene disrupts melanocyte development

The inner ear also has melanocytes! Disrupting these cells causes cell death in hair cells, causing deafness

Inbreeding decreases dog life span

Fig. 3



Mixed breed dogs live an average of 1.2 years longer than purebred dogs

Within purebred Golden retrievers, ones with low inbreeding levels (“outbred”) have higher average lifespans than more highly inbred ones

Yordy et al, Conservation Genet (2020)

Dog breeding and legislature



https://www.npr.org/2024/03/27/1241072309/french-bulldogs-pugs-breed-breathing-health
https://www.npr.org/2022/03/15/1085173405/bulldogs-health-breed-ban
https://www.vettimes.com/news/vets/small-animal-vets/crufts-to-introduce-tougher-brachycephalic-entry-rules

Bulldogs have brachcephaly -- short snouts that impair breathing

They’re also heat-intolerant and get skin infections in their wrinkles

Norway and Denmark have banned their breeding and sale

Breed clubs argue that they can change the breed standard to be healthier

The British Crufts dog show now requires bulldogs pass a respiratory test to be eligible for competition

Dog genetic testing













